

Nicholas W. Landry

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Education

PhD in Applied Mathematics University of Colorado Boulder <i>Advisor: Juan G. Restrepo</i> <i>Dissertation: "Contagion on Complex Systems: Structure and Dynamics"</i>	Boulder, CO 2017–2022
MS in Applied Mathematics University of Colorado Boulder	Boulder, CO 2017–2020
BS in Mechanical Engineering University of New Hampshire <i>University Honors, Summa Cum Laude</i>	Durham, NH 2010–2014

Positions

Academic

Assistant Professor of Biology University of Virginia	Charlottesville, VA August 2024–
Assistant Professor of Data Science (by Courtesy) University of Virginia	Charlottesville, VA January 2025–
External Faculty Vermont Complex Systems Institute, University of Vermont	Burlington, VT August 2024–
TGIR Postdoctoral Research Fellow University of Vermont	Burlington, VT 2022–August 2024

Industry

PhD Intern in the Data Sciences and Analytics Group Pacific Northwest National Laboratory	Seattle, WA Summer 2021
Product Engineer Turbocam International	Barrington, NH 2014–2017

Publications

Journal Articles

- Alyssa Smith, Ilya Amburg, Sagar Kumar, Brooke Foucault Welles, and **Nicholas W. Landry**, *A Blue Start: A large-scale pairwise and higher-order social network dataset*, Scientific Data, 2026. DOI: [10.1038/s41597-026-06920-1](https://doi.org/10.1038/s41597-026-06920-1)

- Martín Coll, Cliff A. Joslyn, **Nicholas W. Landry**, Quintino Francesco Lotito, Audun Myers, Joshua Pickard, Brenda Praggastis, and Przemysław Szufel, *HIF: The hypergraph interchange format for higher-order networks*, Network Science, 2025. DOI: [10.1017/nws.2025.10018](https://doi.org/10.1017/nws.2025.10018)
- Laurent Hébert-Dufresne, Matthew M. Kling, Samuel F. Rosenblatt, Stephanie N. Miller, P. Alexander Burnham, **Nicholas W. Landry**, Nicholas J. Gotelli, and Brian J. McGill, *Stochastic diffusion with approximate master equations with mean-field limits*, Royal Society Open Science, 2025. DOI: [10.1098/rsos.250726](https://doi.org/10.1098/rsos.250726)
- Laurent Hébert-Dufresne, **Nicholas W. Landry**, Juniper Lovato, Jonathan St-Onge, Jean-Gabriel Young, Marie-Ève Couture-Ménard, Stéphane Bernatchez, Catherine Choquette, and Alan A. Cohen, *Governance as a complex, networked, democratic, satisfiability problem*, npj Complexity, 2025. DOI: [10.1038/s44260-025-00041-3](https://doi.org/10.1038/s44260-025-00041-3)
- **Nicholas W. Landry**, Will Thompson, Laurent Hébert-Dufresne, and Jean-Gabriel Young, *Reconstructing networks from simple and complex contagions*, Physical Review E, 2024. DOI: [10.1103/PhysRevE.110.L042301](https://doi.org/10.1103/PhysRevE.110.L042301)
- **Nicholas W. Landry**, Jean-Gabriel Young, and Nicole Eikmeier, *The simpliciality of higher-order networks*, EPJ Data Science, 2024. DOI: [10.1140/epjds/s13688-024-00458-1](https://doi.org/10.1140/epjds/s13688-024-00458-1)
- **Nicholas W. Landry**, Ilya Amburg, Mirah Shi, and Sinan G. Aksoy, *Filtering higher-order datasets*, Journal of Physics: Complexity, 2024. DOI: [10.1088/2632-072X/ad253a](https://doi.org/10.1088/2632-072X/ad253a)
- **Nicholas W. Landry** and Juan G. Restrepo, *Opinion disparity in hypergraphs with community structure*, Physical Review E, 2023. DOI: [10.1103/PhysRevE.108.034311](https://doi.org/10.1103/PhysRevE.108.034311)
- **Nicholas W. Landry**, Maxime Lucas, Iacopo Iacopini, Giovanni Petri, Alice C. Schwarze, Alice Patania, and Leo Torres, *XGI: A Python package for higher-order interaction networks*, Journal of Open Source Software, 2023. DOI: [10.21105/joss.05162](https://doi.org/10.21105/joss.05162)
- **Nicholas W. Landry** and jimi adams, *On limitations of uniplex networks for modeling multiplex contagion*, PLoS ONE, 2023. DOI: [10.1371/journal.pone.0279345](https://doi.org/10.1371/journal.pone.0279345)
- **Nicholas W. Landry** and Juan G. Restrepo, *Hypergraph assortativity: a dynamical systems perspective*, Chaos, 2022. DOI: [10.1063/5.0086905](https://doi.org/10.1063/5.0086905)
- **Nicholas W. Landry**, *Effect of time-dependent infectiousness on epidemic dynamics*, Physical Review E, 2021. DOI: [10.1103/PhysRevE.104.064302](https://doi.org/10.1103/PhysRevE.104.064302)
- **Nicholas W. Landry** and Juan G. Restrepo, *The effect of heterogeneity on hypergraph contagion models*, Chaos, 2020. (Editor's Choice) DOI: [10.1063/5.0020034](https://doi.org/10.1063/5.0020034)
- **Nicholas W. Landry** and Marko Knezevic, *Delineation of First-Order Elastic Property Closures for Hexagonal Metals Using Fast Fourier Transforms*, Materials, 2015. DOI: [10.3390/ma8095303](https://doi.org/10.3390/ma8095303)
- Marko Knezevic and **Nicholas W. Landry**, *Procedures for reducing large datasets of crystal orientations using generalized spherical harmonics*, Mechanics of Materials, 2015. DOI: [10.1016/j.mechmat.2015.04.014](https://doi.org/10.1016/j.mechmat.2015.04.014)

Conference proceedings

- Marko Knezevic, Daniel J. Savage, **Nicholas W. Landry**, *Towards Computationally Tractable Simulations of Metal Forming Processes With Evolving Microstructures*, Proceedings of the ASME International Manufacturing Science and Engineering Conference, 2014. DOI: [10.1115/MSEC2014-3984](https://doi.org/10.1115/MSEC2014-3984)

Book chapters

- Emma Zajdela and **Nicholas W. Landry**, *Higher-order interactions at scientific conferences influence team formation*, Hypergraph Methods in Intelligence Analysis, edited by Hasenjager, M., Fefferman, N. and Bailey, M., 2025. *In press*. [invited chapter] [arXiv:2603.02571](https://arxiv.org/abs/2603.02571)

Preprints

- **Nicholas W. Landry**, Beckett R. Hyde, Jake C. Perez, Sean E. Shaheen, and Juan G. Restrepo, *A theoretical framework for reservoir computing on networks of organic electrochemical transistors*, Preprint, 2024. [arXiv:2408.09223](https://arxiv.org/abs/2408.09223)

Other writings

- **Nicholas W. Landry** and Laurent Hébert-Dufresne, *Group Science: The Open Source Study of Higher-order Networks With XGI*, [SIAM News](#), 2025.

Software

- [Complex Group Interactions](#) (XGI): Creator and Core Developer *NumFOCUS affiliated*
- [HyperContagion](#): Creator and Core Developer
- [HyperNetX](#): Contributor

Funding

- **PI**, [UVA DAC Small Data Analytics Resource Award](#), "Analyzing large-scale social networks through the Bluesky social media platform", 2025-2026
- **Co-writer**, [NSF Award 2309867](#), "Conference: Contagion on Complex Social Systems 2023", (PI: Jean-Gabriel Young, University of Vermont), \$47,838, 2023
- **Co-writer**, [NSF Award 2224051](#), "Conference: Computational Approaches for Contagion on Complex Social Systems", (PI: Juan G. Restrepo, University of Colorado Boulder), \$34,770, 2022
- **Co-writer**, [NSF Award 2121905](#), "HNDS-I: Developing a software library for the analysis and visualization of spreading processes on social hypergraphs", (PI: Juan G. Restrepo, University of Colorado Boulder), \$80,193, 2021-2022

Supervision

Postdoctoral Fellows

- Daniel Kaiser, University of Virginia, 2025-

PhD Students

- Ariana Craft, University of Virginia, 2025-
Co-advised with Tracy Larson
- Abhay Gupta, University of Virginia, 2025-
- Andy Grieve, University of Virginia, 2025-
Primary advisor: Katja Kasimatis
- Charlotte Greene, University of Virginia, 2025-
Primary advisor: Butch Brodie

Master's Students

- Will Thompson, University of Vermont, 2023-2024
- Erik Weis, University of Vermont, 2022-2023

Undergraduate Students

- Rudra Dave, University of Virginia, 2025-
- Tyler Long, University of Virginia, 2025-
- Gelila Solomon, University of Virginia, 2025-
- Ahmed Ahmed, University of Virginia, 2025
Co-mentored with Baltazar Espinoza
- Yifei Luo, Middlebury College, 2024-2025
Co-mentored with Phil Chodrow
- Adeline Southard, University of Vermont, 2024
Co-mentored with Jean-Gabriel Young
- Beckett Hyde, University of Colorado Boulder, 2022-2024
Co-mentored with Juan G. Restrepo
- Emerson McMullen, Harvey Mudd College, 2022
Co-mentored with Juan G. Restrepo and Heather Zinn Brooks
- Arjun Asija, Harvey Mudd College, 2022
Co-mentored with Juan G. Restrepo and Heather Zinn Brooks

Rotation students

- Michael Leonard, University of Virginia, 2026
- Kuankuan Hu, University of Virginia, 2025

Presented work

Invited talks

- *Reconstructing networks and dynamics from contagion data* November 2026
ASA Section on Statistical Learning and Data Science 2026 New York, NY
- *Reconstructing networks and dynamics from contagion data* May 2026
Department of Genome Sciences at the University of Virginia Charlottesville, VA
- *A practical guide to higher-order networks* November 2025
Datapalooza at UVA School of Data Science Charlottesville, VA
- *Opinion disparity in hypergraphs with community structure: theory and practice* May 2025
SIAM Dynamical Systems Denver, CO
- *Realistically modeling diseases: From data to models and back again* March 2025
Bryn Mawr College Bi-Co Mathematics Colloquium Bryn Mawr, PA
- *Realistically modeling diseases: From data to models and back again* February 2025
University of Virginia Statistics Colloquium Charlottesville, VA
- *Realistically modeling diseases: From data to models and back again* November 2024
Grinnell College Grinnell, IA
- *Opinion disparity in hypergraphs with community structure: theory and practice* October 2024
DIMACS Workshop at Rutgers University New Brunswick, NJ
- *Modeling contagion processes with noisy and uncertain networks* October 2024
Graph and Network Data Seminar at the University of Virginia Charlottesville, VA
- *Modeling contagion processes with noisy and uncertain networks* October 2024
EEBio Seminar at the University of Virginia Charlottesville, VA
- *Reconstructing networks from complex social processes* September 2024
Quantitative Collaborative Seminar at the University of Virginia Charlottesville, VA

- *Complex Group Interactions (XGI)* June 2024
Software Tools for Network Science Satellite at NetSci Quebec City, Quebec, Canada
- *Realistically modeling diseases: From data to models and back again* April 2024
WINQ Program on Complex and Quantum Systems Stockholm, Sweden
- *Higher-order structure is more complex than current measures and models* April 2024
Network Seminar Series of the CRI, LPI Paris
- *Modeling contagion processes with higher-order networks* February 2024
University of Virginia
- *Modeling contagion processes with higher-order networks* January 2024
Worcester Polytechnic Institute
- *Modeling contagion processes with higher-order networks* January 2024
University at Buffalo
- *Limitations and opportunities from simple higher-order structural and contagion models*
September 2023
Vermont-KIAS Workshop: Group Interactions in Network Science Burlington, VT
- *Higher-order interaction networks: structure, dynamics, and inference* May 2023
Workshop on Modelling and Mining Complex Networks as Hypergraphs Toronto, Canada
- *Higher-order models for social and epidemiological contagion* January 2023
Network Science Institute at Northeastern Boston, MA
- *Community structure in hypergraphs and the emergence of polarization* October 2022
AMS Fall Eastern Sectional Meeting Amherst, MA
- *Hypergraph dynamics: assortativity and the expansion eigenvalue* April 2022
Joint Mathematics Meetings
- *Hypergraph assortativity: A dynamical systems perspective* March 2022
APS March Meeting
- *Contagion on Complex Systems: Structure and Dynamics* January 2022
Harvard Center for Communicable Disease Dynamics
- *Contagion on Complex Systems: Structure and Dynamics* January 2022
University of Vermont
- *Contagion on Complex Systems: Structure and Dynamics* January 2022
Dartmouth College
- *Contagion on Complex Systems: Structure and Dynamics* January 2022
CU Boulder Applied Mathematics Dynamics Seminar
- *Hypergraph dynamics: a dynamical systems perspective* December 2021
Graph Theory and its Applications session at the 2021 Winter Canadian Mathematical Society (CMS) Meeting
- *The effect of contact structure on hypergraph contagion models* May 2021
Dynamics on Networks with Higher Order Interactions Minisymposium, SIAM Dynamical Systems Conference
- *The effect of heterogeneity on hypergraph contagion models* October 2020
Fundamentos y Enseñanza de la Física y los Sistemas Dinámicos, Universidad de Antioquia
- *The effect of heterogeneity on hypergraph contagion models* September 2020
CU Boulder Applied Mathematics Dynamics Seminar
- *Hypergraph Contagion* February 2020
Colorado Chapter of Society of Young Network Scientists

Contributed talks

- *Efficient sampling from the hypergraph configuration model* June 2025
NetSci Maastricht, the Netherlands
- *Governance as a complex, networked, democratic, satisfiability problem* March 2025
APS Global Physics Summit Anaheim, CA
- *Nonparametric approach to network reconstruction from time-series data* June 2024
NetSci Quebec City, Quebec, Canada
- *Learnability of complex structure from contagion of various complexities* March 2024
APS March Meeting Minneapolis, MN
- *XGI: A Python package for higher-order interaction networks* July 2023
NetSci Vienna, Austria
- *Hypergraph community structure and the emergence of polarization* October 2022
Conference on Complex Systems Palma, Spain
- *Hypergraph community structure and the emergence of polarization* September 2022
SIAM Network Science Workshop
- *Hypergraph community structure and the emergence of polarization* July 2022
NetSci
- *Hypergraph community structure and the emergence of polarization* March 2022
Northeast Regional Conference on Complex Systems (Best Oral Presentation)
- *Hypergraph dynamics: assortativity and the expansion eigenvalue* November 2021
International Conference on Complex Networks and their Applications
- *On limitations of uniplex networks for modeling multiplex diffusion* July 2021
Networks
- *Hypergraph community structure and the emergence of polarization* June 2021
TopoNets: Networks Satellite
- *The effect of time-dependent infectiousness on epidemic dynamics* March 2021
Front Range Applied Mathematics Student Conference
- *The effect of heterogeneity on hypergraph contagion models* September 2020
TopoNets: NetSci Satellite Conference
- *Improvisatory Elements of Teaching* February 2019
Workshop for the Graduate Teacher Program Boulder, CO
- *So You Think You're Bad at Math* January 2019
Ignite Talk for the Graduate Teacher Program's Spring Conference Boulder, CO
- *Music Data Mining: Finding Structure in Song* Fall 2018
Statistics, Optimization, and Machine Learning Seminar, Applied Math Boulder, CO

Posters

- *Reconstructing networks from simple and complex contagions* January 2025
Dynamics Days
- *Community structure in hypergraphs and the emergence of polarization* January 2022
Dynamics Days
- *The effect of time-dependent infectiousness on epidemic dynamics* March 2021
Northeastern Regional Conference on Complex Systems
- *The effect of heterogeneity on hypergraph contagion models* August 2020
Dynamics Days Digital
- *The effect of simplex and network degree distribution on simplicial contagion models* January 2020

Dynamics Days

Hartford, CT

Tutorials

- *Minitutorial: A Practical Guide to Modeling with Higher-order Networks* May 2025
SIAM Conference on Applications of Dynamical Systems Denver, CO
- *GSNP Short Course on Higher Order Network Science* March 2024
APS March Meeting Minneapolis, MN

Software demonstrations

- *XGI* May 2025
Conference on Applications of Dynamical Systems Denver, CO
- *XGI* May 2023
Workshop on Modelling and Mining Complex Networks as Hypergraphs Toronto, Canada
- *XGI* October 2022
TopoNets Satellite Conference of the Conference on Complex Systems Palma, Spain
- *XGI* July 2022
Higher-Order Models in Network Science Satellite Conference of NetSci Online
- *XGI and HyperContagion* August 2022
Contagion on Complex Social Systems Workshop Boulder, CO

Teaching

Experience

Instructor

Charlottesville, VA

University of Virginia

2025-Present

- Advanced Ecology & Evolution 2 (BIOL 8082/8084): Spring 2026
- Networks in Biology: From Pandemics to Ecosystems (BIOL 4559): Fall 2025

Instructor

Boulder, CO

University of Colorado Boulder

2020

- Calculus 1 for Engineers (APPM 1350): Summer 2020

Teaching Assistant

Boulder, CO

University of Colorado Boulder

2017-2021

- Calculus 1 for Engineers (APPM 1350): Fall 2017
- Calculus 2 for Engineers (APPM 1360): Spring 2018, Summer 2019, Fall 2019
- Calculus 3 for Engineers (APPM 2350): Fall 2018
- Differential Equations and Linear Algebra (APPM 2360): Spring 2019, Fall 2020, Spring 2021
- Matrix Methods (APPM 3310): Spring 2020

Professional Development

Center for Teaching Excellence

Charlottesville, VA

University of Virginia

- Ignite Fellowship: September 2025 - December 2025
- Course Design Institute: May 2025

Graduate Teacher Program

Boulder, CO

Certificate in College Teaching

November 2018

- Attended 20 hours of teaching-related workshops
- Observed by a faculty member to vouch for my teaching
- Participated in 2 consultations using video footage from my class
- Attended 20 hours of discipline-specific teaching workshops.
- Wrote a teaching portfolio, outlining my teaching experience, skills, and philosophy

Awards

- Zachary Karate Club Award 2024
- Chief Student Marshal for UNH Commencement 2014 based on GPA and contributions to the college 2014
- Mechanical Engineering Faculty Choice Award for Poster at UNH Undergraduate Research Conference 2014
- Nominee for the Goldwater Scholarship; 1 of 4 students representing UNH 2012
- Eagle Scout 2008

Service

PhD committees

- Charlotte Greene
- Andy Grieve

DMP second reader

- Reece Anderson, 2025
- Griffin Jiron, 2025

Leadership

- Graduate Peer Mentor, University of Colorado Boulder, 2020-2021
- Lead Teaching Assistant, University of Colorado Boulder, 2018-2019
- Graduate Student Representative, University of Colorado Boulder, 2018-2019
- Tutor, I Have a Dream Foundation of Boulder County, 2018
- Vice President of UNH Chapter of Pi Mu Epsilon, University of New Hampshire, 2012-2013

Conferences and seminars organized

Co-chair

Contagion on Complex Social Systems Workshop (CCSS)

- Burlington, VT, August 14-16, 2023
- Boulder, CO, August 10-12, 2022

Center for Teaching Excellence

Charlottesville, VA

University of Virginia

- Teaching Triangles: February 2026 - April 2026
- Ignite Fellowship: September 2025 - December 2025
- Course Design Institute: May 2025

Co-chair

Software and Data for Supporting Network Science satellite workshop

- Maastricht, Netherlands, June 2, 2025 (Co-located with NetSci)

Co-organizer

TopoNets satellite workshop

- Maastricht, Netherlands, June 2, 2025 (Co-located with NetSci)
- Quebec City, Canada, June 17, 2024 (Co-located with NetSci)
- Vienna, Austria, July 10, 2023 (Co-located with NetSci)
- Palma, Spain, October 20, 2022 (Co-located with CCS)

Co-organizer

TalkbOctopus seminar series

Burlington, VT
Fall 2022 - Spring 2023

Co-organizer

Models and Methods for Sparse (Hyper) Network Science

- Boston, MA, January 6, 2023 (Co-located with JMM)

Joint coordinator of the Dynamical Systems seminar

CU Boulder Applied Math Department

Boulder, CO
Spring 2021, 2022

Program committees

- The International School and Conference on Network Science (NetSci): 2024-2026
- Workshop on Modelling and Mining Networks (WAW): 2024-2026

Peer review

Journals

BMC Bioinformatics; Chaos; Chaos, Solitons, and Fractals; Journal of Epidemiology and Global Health; Journal of Machine Learning Research; Journal of Open Source Software; Journal of Physics: Complexity; Journal of Statistical Physics; Mathematical Biosciences; Nature Communications; Nature Communication Physics; npj Complexity; Physical Review E; Physical Review Letters; Physical Review Research; PLoS Complex Systems; PLoS ONE; Science Advances; Scientific Reports

Conferences

Algorithm Engineering and Experiments (2022)

Other professional activities

Workshops attended

- *The Course Design Institute at the University of Virginia* May 2025
Participant Charlottesville, VA
- *Workshop on Spreading on Social Networks at DIMACS at Rutgers* October 2024
Participant New Brunswick, NJ
- *WINQ Program on Complex and Quantum Systems* April 2024
Participant Stockholm, Sweden
- *Complex Networks Winter Workshop* December 2023
Participant Quebec City, Quebec, Canada
- *MRC: Complex Social Systems* June 2023
Participant Buffalo, NY

- *Modeling Pandemic Intervention Acceptance for Disease Mitigation* April 2023
Participant Online
- *JSMF-SFI Postdocs in Complexity Conference X* March 2023
Participant Santa Fe, NM
- *MRC: Models and Methods for Sparse (Hyper) Network Science* June 2022
Participant Buffalo, NY
- *Complex Networks Winter Workshop (CNWW)* January 2021
Participant Online
- *Statistics and Modeling with Novel Data Streams at the SISIMID summer school* June 2020
Participant Online
- *Understanding and Exploring Network Epidemiology in the Time of Coronavirus* April 2020
Participant Online

Media

- [*Reconstructing networks from simple and complex contagions*](#)
October 28th, 2024
Quantum Photonics Club podcast
- [*Are Ideas Contagious?*](#)
October 9th, 2024
University of Virginia College of Arts & Sciences Press Release
- [*Interactions Within Larger Social Groups Can Cause Tipping Points in Contagion Flow*](#)
October 20th, 2020
AIP Press Release
- [*Contagion on Complex Networks*](#)
February 3rd, 2020
Radio, Season 3 Episode 13, Probably Novel at University of Colorado Boulder

Travel Grants

- *CU Boulder Graduate School Student Travel Grant* 2020, 2022
- *2022 JMM Grad Student Travel Grant* 2022
Awarded a \$1,300 travel grant
- *Networks 2021 Registration Waiver* 2021
Awarded a registration waiver for Networks 2021 which is being held virtually
- *SIAM Student Travel Award* 2021
Awarded a registration waiver for SIAM DS 2021 which is being held virtually